PHYS4055 – Atomic Physics and Quantum Optics – Semester 2, 2026
Assignment 3
Deadline: 4pm Friday 25 September 2026

This assessment contributes 25%/4 towards your overall grade. It consists of *four* questions on topics covered in the first few weeks of the atomic physics part of the course. Each question carries equal weight. In answering these questions *show all working*. While you may consult the reference texts and discuss the questions broadly with your peers, you need to carry out and submit *your own work*.

1. Nuclear-size effects.

   In class, we have explored the first-order corrections to the energies in hydrogen and H-like ions arising from relativistic and spin-orbit effects. So far, we have treated the nucleus as point-like. In this problem, we are going to explore the effect of account of finite nuclear size on the energies.

   Let's model the charge distribution of the nucleus as uniform within a radius R, corresponding to the nuclear charge radius. This corresponds to a step-function density profile, and is a reasonable model. The nuclear charge radius R is related to the root-mean-square charge radius $R_{\rm rms}$ – which may be found in data tables – as $R = \sqrt{5/3}\, R_{\rm rms}$.

   This nuclear charge distribution corresponds to the following expression for the nuclear potential inside the radius R:

   $$V(r) = -\frac{Z}{R}\left[\frac{3}{2} - \frac{1}{2}\left(\frac{r}{R}\right)^2\right] , \tag{1}$$

   and outside the nucleus ($r \geq R$) the potential is $V(r) = -Z/r$.

   (a) Show that the leading correction (in powers of R) for s-states is

   $$\Delta E_{\rm fns} = \frac{2Z^2}{5n^3}(ZR)^2 . \tag{2}$$

   (b) Using the Mathematica codes, or your favourite programming language, numerically evaluate the finite-nuclear-size energy corrections for $1s$, $2p$, and $3d$ states of H-like ^{209}Bi. Submit your code and include comments. How does your numerical result compare to what you obtained analytically for the s-states?

   (c) Compare your results from (b) with the size of the relativistic corrections we considered in class for the same system and states. Comment on your results.

2. Electron configurations and terms.

   Coupling of two identical d states.

   (a) Find all possible terms from the configuration nd^2, where n is the principal quantum number, showing all working.

   (b) What is the expected ordering of the energy levels of these terms? Explain your answer.

   (c) Peruse the Periodic Table and pick your favourite element with an unpaired nd^2 in the ground state, with all other shells or subshells filled (e.g., $n's^2$, $n'f^{14}$). Consulting the NIST Atomic Spectra Database, do the energy levels follow the ordering you predicted for the terms? Comment on the similarities/differences.

   (d) For *one* of the considered excited states (terms) from your answer to (c), find the *excitation energy* (energy above the ground state) in both atomic units and cm^{-1}.

3. Atomic polarisability.

   Due to the interaction of a static electric field $\boldsymbol{\varepsilon}$ with an atom or ion, the atomic energy levels may shift (Stark shift). The interaction term is given by $H_E = -\sum_i \mathbf{d}_i \cdot \boldsymbol{\varepsilon} = e\sum_i z_i \varepsilon = e\sum_i \varepsilon r_i \cos\theta$, where the electric field direction defines the z axis, and ε is uniform. The energy shift ΔE of the state n arises in the second order of perturbation theory,

$$\Delta E = \sum_m \frac{\left|\langle\psi_n|H_E|\psi_m\rangle\right|^2}{E_n - E_m} = -\frac{1}{2}\alpha\varepsilon^2 \ ,$$

   where relevant quantum numbers are wrapped up in the state labels, and the sum runs over all possible states that satisfy the angular momentum and parity rules. This energy shift is often expressed in terms of the atomic *polarisability* α, which gives a measure of how polarisable or distorted the atom is on application of an electric field.

   Here, we will consider the simple case of the Stark shift for a *hydrogen-like ion.*

   (a) Approximating the energy denominator $E_1 - E_m$ by $E_1 - E_2$, where E_1 and E_2 are the unperturbed ground and first-excited state energies, calculate the polarizability α in the ground state of a hydrogen-like ion, in terms of Z.

   [*Hint*: You may wish to use *closure*, $\sum_m |\psi_m\rangle\langle\psi_m| = 1$, to simplify for your expression.]

   (b) From this result, determine a numerical value for the polarisability for hydrogen (Z=1), expressing your result in both atomic units a_B^3 and cm^3, where a_B is the Bohr radius. Compare your value to the exact value for hydrogen $1s$, $\alpha = 9/2\, a_B^3$, and comment on how your value compares.

   (c) Similarly, determine a numerical value for hydrogen-like rubidium $1s$ (Z=37), again expressing your result in both atomic units and cm^3. Explain why this value is significantly different to that for the ground state of neutral rubidum, $319\, a_B^3$.

4. Landé factor.

   In this problem we will consider the interaction of an external magnetic field with the magnetic moment of an atom, $H_B = -\boldsymbol{\mu} \cdot \mathbf{B}$.

   We will take the magnetic moment to be comprised of orbital and spin parts of the electrons, $\boldsymbol{\mu} = -\mu_B(g_L\mathbf{L} + g_S\mathbf{S})$, where μ_B is the Bohr magneton, and $g_L = 1$ and $g_S \approx 2$ are the orbital and spin g-factors of the electron. For simplicity, we will take $g_S = 2$. We will ignore the smaller contribution from the nuclear magnetic moment.

   (a) Considering the total angular momentum of the system to be $\mathbf{J} = \mathbf{L} + \mathbf{S}$, and hence $\boldsymbol{\mu} = -g_J\mu_B\mathbf{J}$, show that the so-called Landé factor g_J is given by

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \ .$$

   (b) Now consider the case where the total angular momentum of the atom $\mathbf{F} = \mathbf{I} + \mathbf{J}$ is considered, hence $\boldsymbol{\mu} = -g_F\mu_B\mathbf{F}$, and with the nuclear magnetic moment contribution still neglected. Find the corresponding expression for the Landé factor g_F.

   (c) Consider the term $^2P_{1/2}$. Considering the case with total angular momentum $\mathbf{F}$, explain how this term splits on applicaton of a magnetic field. Obtain the possible values for g_F in terms of nuclear spin I.